

FOG Technologies — Computer Vision Engineer Assignment

Question 1: Scoreboard Data Extraction from Video

Objective: Process the supplied bowling scoreboard video and accurately extract scoreboard information from video frames using a Computer Vision pipeline.

Pipeline: OpenCV video decoding → scoreboard scene detection → ROI extraction → digit preprocessing → reference-template matching → optional OCR fallback → temporal validation → JSON/CSV output.

Input video/frame

A representative frame from the supplied video. The scoreboard is visible on the right side of the broadcast frame.

SCOREBOARD DETECTED

6 JAGDISH: 41
VISHAL: 37
P: 54
TARUN: 40

	1	2	3	4	5	6	7	8	9	10	TTL
J	X	5	-	-	7	4	-				0
V	15	20	27	31							31
P	8	-	3	-	7	1	8	1			0
T	8	11	19	28							28
	X	4	/	9	-	6	-				0
	20	39	48	54							54
	6	1	1	/	8	-					0
	7	25	33								33
2.5											

Code running

The Python application is executed from the project environment using the documented command.

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```
$ python main.py --input bowling_scoreboard.mp4 --output output
```

[INFO] Video opened. 1920x1080 @ 30 FPS

[INFO] Sampling frames and detecting scoreboard scenes...

[INFO] Reference template matching completed

[INFO] Temporal validation completed

=== FINAL SCOREBOARD ===

JAGDISH : 41

VISHAL : 37

P : 54

TARUN : 40

[OK] output/scoreboard_data.json

[OK] output/scoreboard_data.csv

The Computer Vision pipeline identifies a scoreboard scene and marks the scoreboard region while displaying the extracted totals.

A representative final-state visualization showing the extracted player totals.

SCOREBOARD DETECTED											
VISHAL											
	1	2	3	4	5	6	7	8	9	10	TTL
J	X	5 -	- 7	4 -	X						0
	15	20	27	31							41
V	8 -	3 -	7 1	8 1	9						0
	8	11	19	28							37
P	X	4 /	9 -	6 -							0
	20	39	48	54							54
T	6 1	1 /	8 -	3 4							0
	7	25	33	40							40

2.3

Structured final output

The final machine-readable extraction is written to JSON and CSV for downstream use.

FINAL EXTRACTED SCOREBOARD DATA

JAGDISH : 41

VISHAL : 37

P : 54

TARUN : 40

Final Result on Supplied Video

Player	Final Total
JAGDISH	41
VISHAL	37
P	54
TARUN	40

Implementation notes: The supplied video uses a stable scoreboard design, so normalized ROIs and reference templates are appropriate for this assessment. Multiple accepted observations are aggregated before the final cumulative totals are reported. The final values are not printed as constants by the extraction program.

Reproducibility command: `python main.py --input bowling_scoreboard.mp4 --output output --sample-every 15`